

Primene paralelnog računarstva

Primene paralelnog računarstva

- **Složena i obimna izračunavanja**
- **Obrada velike količine podataka** (oblasti nauke o podacima (engl. *data science*) i sistema za rad sa velikim količinama podataka (engl. *big data*))
 - Internet (veb pretraživači), genetika (DNK), medicinske i astronomske slike, senzori (engl. *Internet-of-Things*, IoT)
- **Simulacija**
- **Vizuelizacija**
- Svi prethodno opisani problemi zahtevaju znatne resurse za izračunavanja, vode nas do oblasti **računarstva visokih performansi** (engl. *High Performance Computing*)
 - Klasteri, data centri, super-računari
 - Programiranje super-računara: ssh protokol za komunikaciju, MPI + OpenMP/CUDA/OpenACC

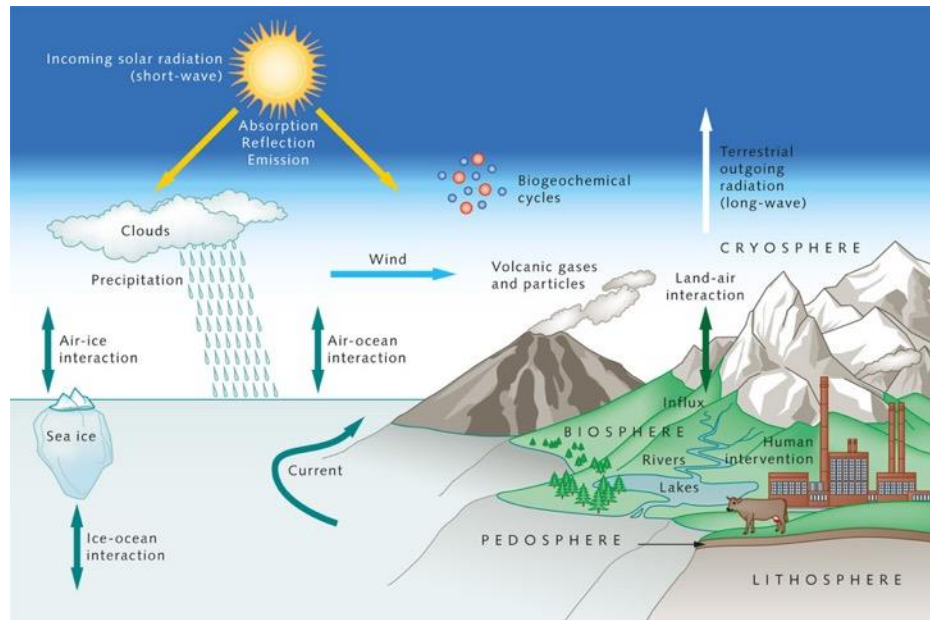
Primene paralelnog računarstva

- **Naučna izračunavanja** (engl. *scientific computing*)
 - Klima i vremenska prognoza, genetika, astrofizika, okeanografija, seizmologija, farmacija, ekonomija i finansije, ...
- **Veštačka inteligencija** (engl. *artificial intelligence*)
 - Mašinsko učenje, duboke neuronske mreže, ...
- **Inženjerstvo**
 - Projektovanje vozila, letelica i mašina, razvoj novih materijala, ...
- **Energetika**
 - Obnovljivi izvori energije, istraživanje tla , ...
- **Medicina**
 - Obrada medicinskih slika, ...



Naučna izračunavanja – vreme i klima

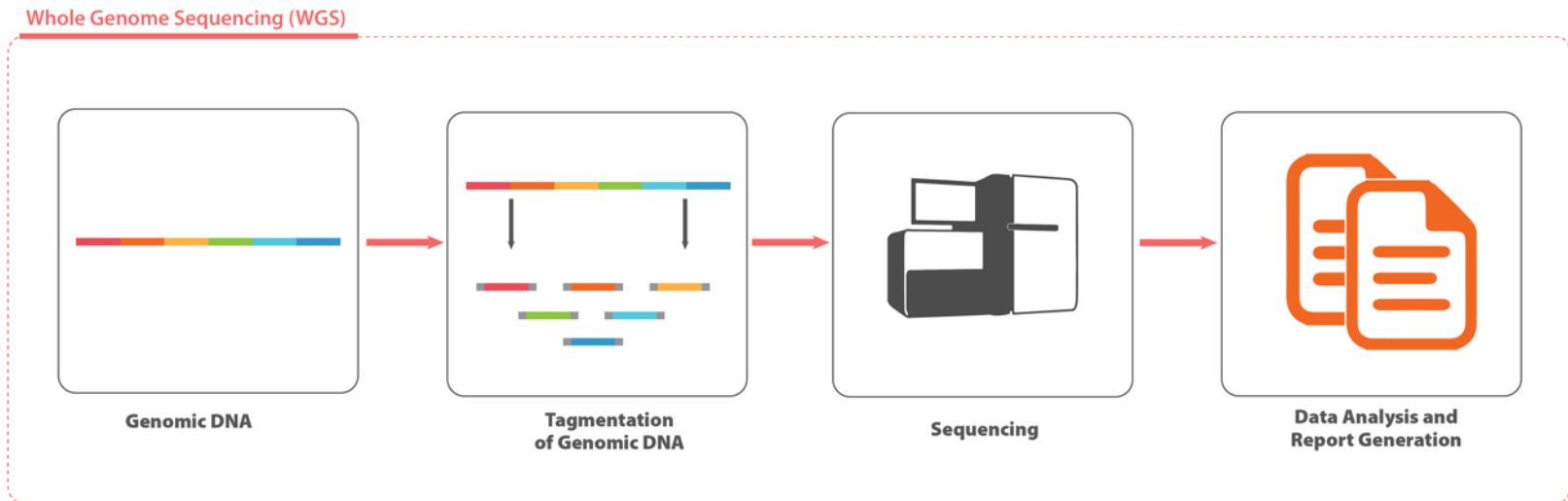
- **Tačnost prognoze vremena i klime** veoma zavisi od **rezolucije i kompleksnosti** korišćenih modela
- **Zemljina atmosfera** predstavlja **nelinearni dinamički sistem** podložan **haosu**
- Simulacije na osnovu ovakvih modela zahtevaju **znatne računarske resurse**



Izvor: <https://worldoceanreview.com/en/world-climate-system/earth-climate-system/>

Naučna izračunavanja – genetika

- **Sekvenciranje genoma** podrazumeva vrlo kompleksna izračunavanja (algoritmi za poklapanje, edit distance i rad sa stringovima), Human Genome Project doveo do sekvenciranja ljudskog genoma 2001.
- **Napredak paralelnog računarstva** omogućio snižavanje cene sekvenciranja
- Osnova za razvoj bioinformatike, novih lekova i **personalizovane medicine**



Izvor: <https://www.abmgood.com/Whole-Genome-Sequencing-Service.html>

Naučna izračunavanja – ekonomija

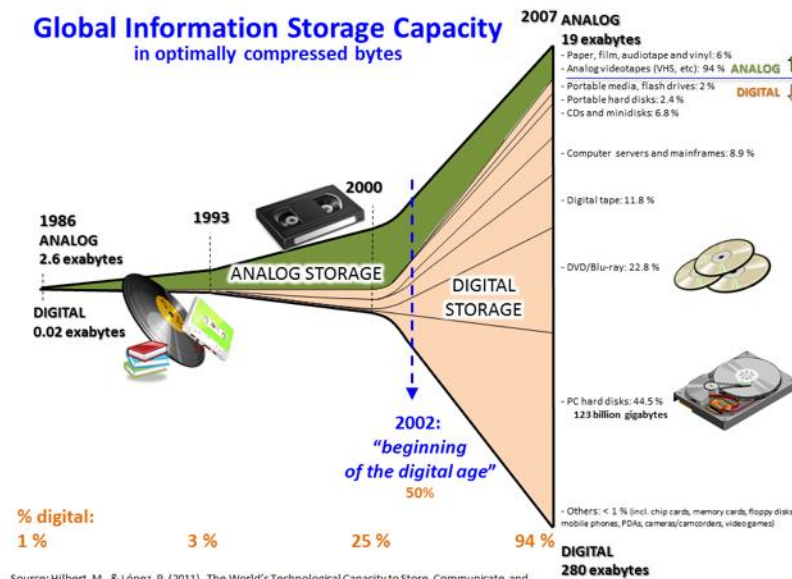
- **Analizom finansijskih podataka** koji se danas generišu velikom brzinom može se doći do **novih znanja** na osnovu kojih se mogu donositi **bolje odluke**, želja je imati **pravu informaciju u realnom vremenu**
- **Paralelizacija** donosi unapređenja kod algoritama u oblastima kao što su **određivanje cene finansijskih instrumenata, upravljanje rizicima i simulacija tržišta**
- Za rešavanje problema u oblasti ekonomije i finansija, danas se koristi **kombinacija heterogenog paralelnog i distribuiranog računarstva**



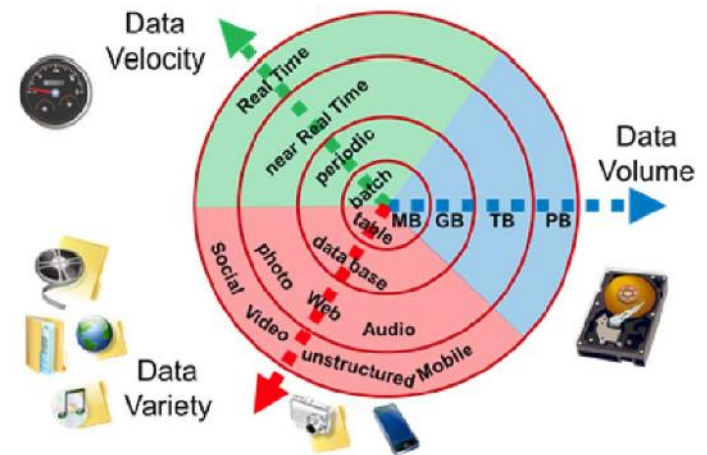
Izvor: <https://www.fool.com.au/2020/02/03/oil-search-share-price-drops-7-on-png-decision/>

Obrada velikih skupova podataka

- Količina podataka koja se čuva u svetu duplira se otprilike na svake dve godine, veličina „digitalnog univerzuma“ u 2020. je oko 44 ZB (10^{21} bajta), ako nisu predmet analize, skoro svi ti podaci su beskorisni
- Oblasti nauke o podacima i sistema za rad sa velikim količinama podataka se u velikoj meri oslanjaju na paralelnu i distribuiranu obradu



Source: Hilbert, M., & López, P. (2011). The World's Technological Capacity to Store, Communicate, and Compute Information. *Science*, 332(6025), 60–65. <http://www.martinhilbert.net/WorldInfoCapacity.html>



Izvor: https://en.wikipedia.org/wiki/Big_data

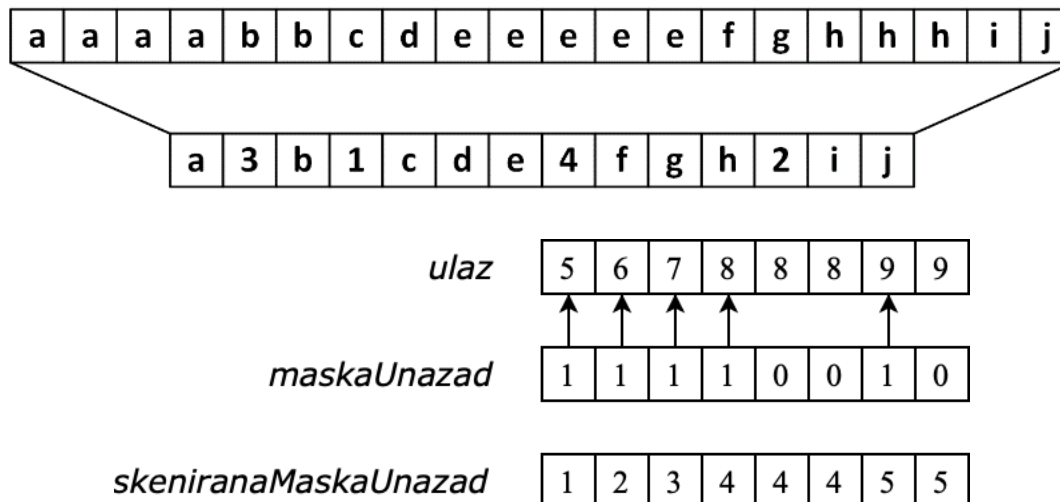
Obrada velikih skupova podataka

- Često korišćena strategija za obradu velikih skupova podataka podelom na delove koji se mogu paralelno i distribuirano obrađivati jeste obrazac **mapiranje/redukcija**
- Prvo se vrši **mapiranje** (filtriranje, sortiranje), potom ide **redukcija** (sumiranje), **stablo redukcije** (engl. *reduction tree*) se koristi kako bi se sumirali rezultati za svaki od delova u konačni odgovor
- **MapReduce** je programski model koji predstavlja primer implementacije ovog obrasca – open source projekat Apache Hadoop
- Na GPU, svaka nit može da obradi deo podataka i pritom koristi deljenu memoriju za čuvanje međurezultata, brzina rada **atomičnih operacija**, koje su ključne za **efikasnu redukciju**, je značajno unapređena u novijim GPU arhitekturama



Primer – Kompresija podataka

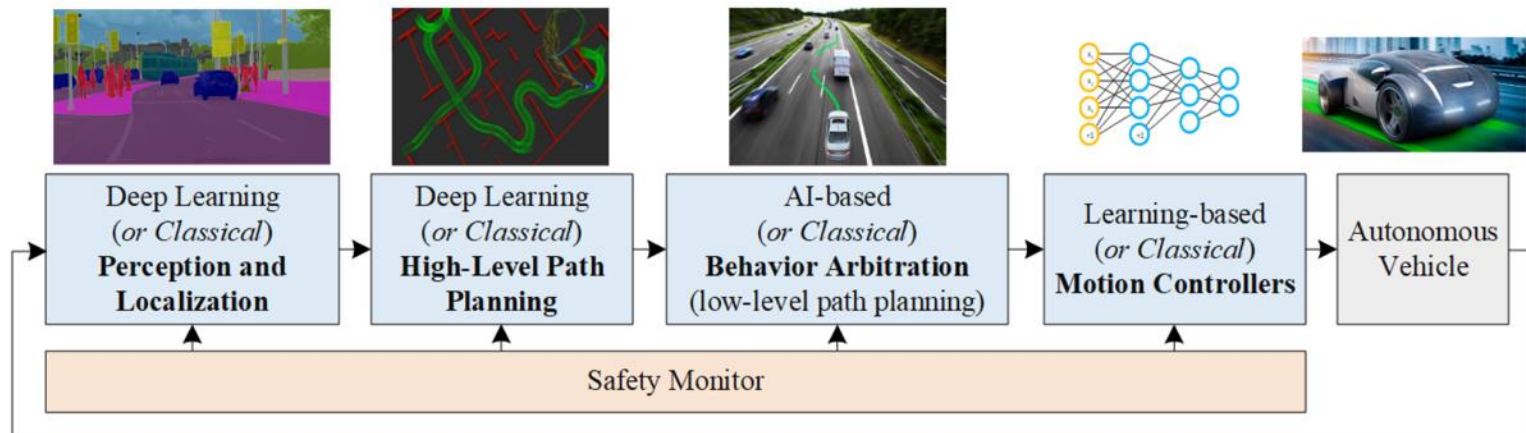
- Kompresiona tehnika **ran-lengt** (engl. *run-length encoding* – RLE) može se efikasno paralelizovati primenom obrasca **skeniranja** (engl. *scan*) – PARLE algoritam
- Skeniranjem se dobijaju izlazne pozicije svih znakova i time omogućava paralelizacija do nivoa da svaki od elemenata računa zasebna nit, veoma pogodna organizacija izračunavanja za visoko-paralelnu GPU arhitekturu



Izvor: Dragan, Gajić, "Uvod u kompresiju podataka – osnovni pojmovi i entropijske kompresione tehnike"

Mašinsko učenje

- **Duboko učenje** (engl. *deep learning*) zasniva se na visoko-paralelnoj obradi na grafičkim procesorima
- **Duboke neuronske mreže** imaju veći broj slojeva između ulaznog i izlaznog (rekurentne, konvolucione, transformeri...)
- **Primene:** prepoznavanje slike i govora, bioinformatika, obrada jezika, sistemi za preporučivanje, autonomna vožnja (Nvidia DRIVE OS, DriveWorks SDK)



Izvor: <https://arxiv.org/pdf/1910.07738.pdf>

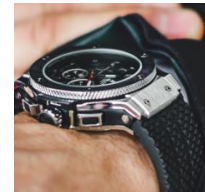
Primer – Duboko učenje na Nvidia DGX



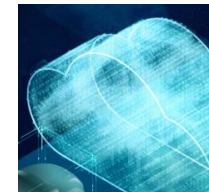
POWERED BY 4 NVIDIA TESLA V100 GPUs
BUILT ON THE LATEST NVIDIA VOLTA™ GPU ARCHITECTURE



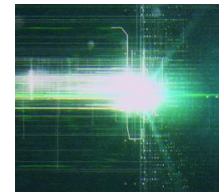
The power of
500 TFLOPS of AI Power



Get started
within one hour



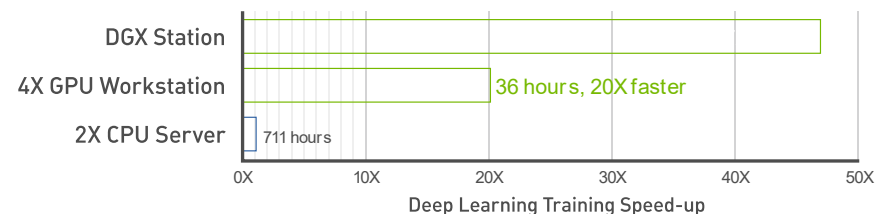
No data center?
No problem.



Access to
AI expertise

ITERATE AND INNOVATE FASTER
UNPARALLELED DEEP LEARNING TRAINING PERFORMANCE

NVIDIA DGX Station Delivers 47X Faster Training



DGX Station performance projected based on DGX (with Tesla V100) Workload: ResNet50, 90 epochs to solution | CPU Server: Dual Xeon E5-2699 v4, 2.6 GHz. Projections subject to change.

Izvor: <https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/dgx-station/dgx-station-print-Infographic-738375-nvidia-web.pdf>

Primer – Duboko učenje na Nvidia DGX

Softver

CONTAINERIZED APPLICATION

DEEP LEARNING APPLICATIONS
DEEP LEARNING FRAMEWORKS
DEEP LEARNING LIBRARIES
CUDA TOOLKIT

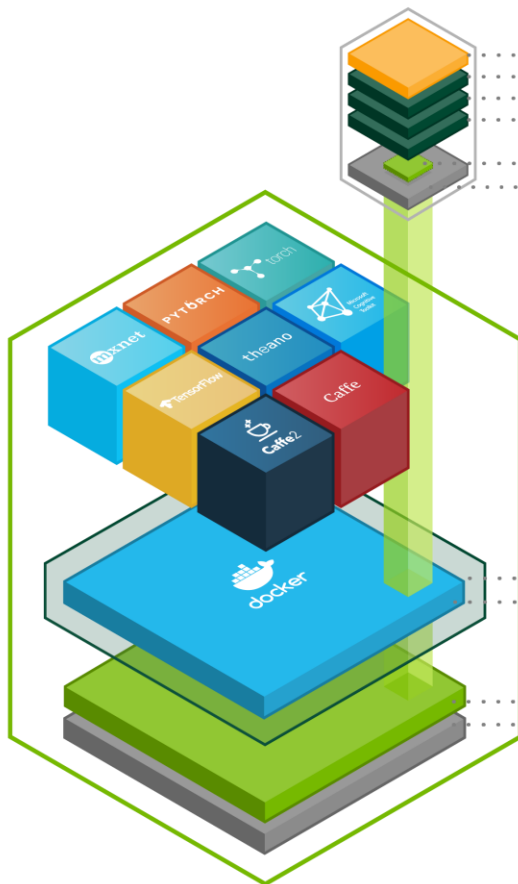
MAPPED NVIDIA DRIVER
CONTAINER OS

CONTAINERIZATION TOOL

NVIDIA CONTAINER RUNTIME FOR DOCKER
DOCKER ENGINE

NVIDIA DRIVER
HOST OS

NVIDIA DGX SOFTWARE STACK



Hardver

1. GPUs

4X NVIDIA Tesla® V100 32 GB/GPU
500 TFLOPS (Mixed Precision)
20,480 Total NVIDIA CUDA® Cores
2,560 Tensor Cores

2. SYSTEM MEMORY

256 GB RDIMM DDR4

3. GPU INTERCONNECT

NVIDIA NVLink™,
Fully Connected 4-Way

4. STORAGE

Data: 3 x 1.92 TB SSD RAID 0
OS: 1 x 1.92 TB SSD

5. CPU

Intel Xeon E5-2698 v4
2.2 GHz 20-Core

6. NETWORKING

2X 10 GbE

7. DISPLAYS

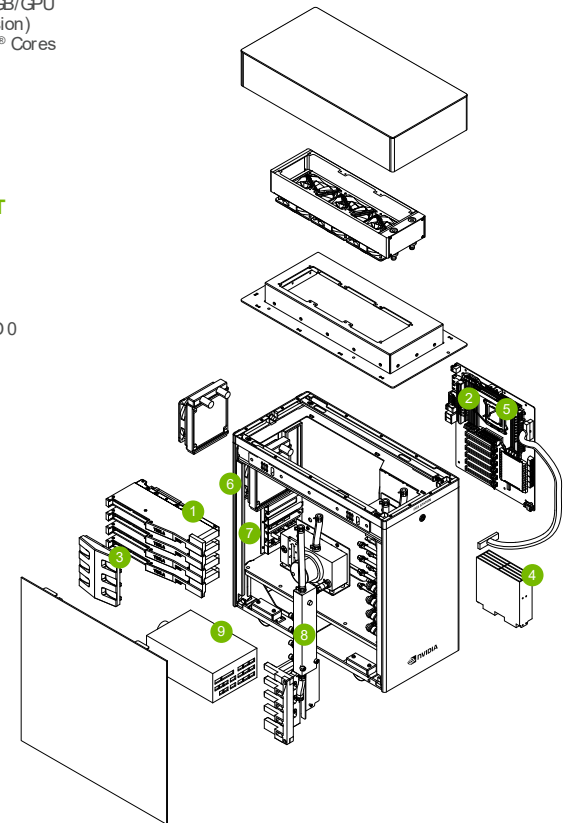
3X DisplayPort,
4K Resolution

8. COOLING

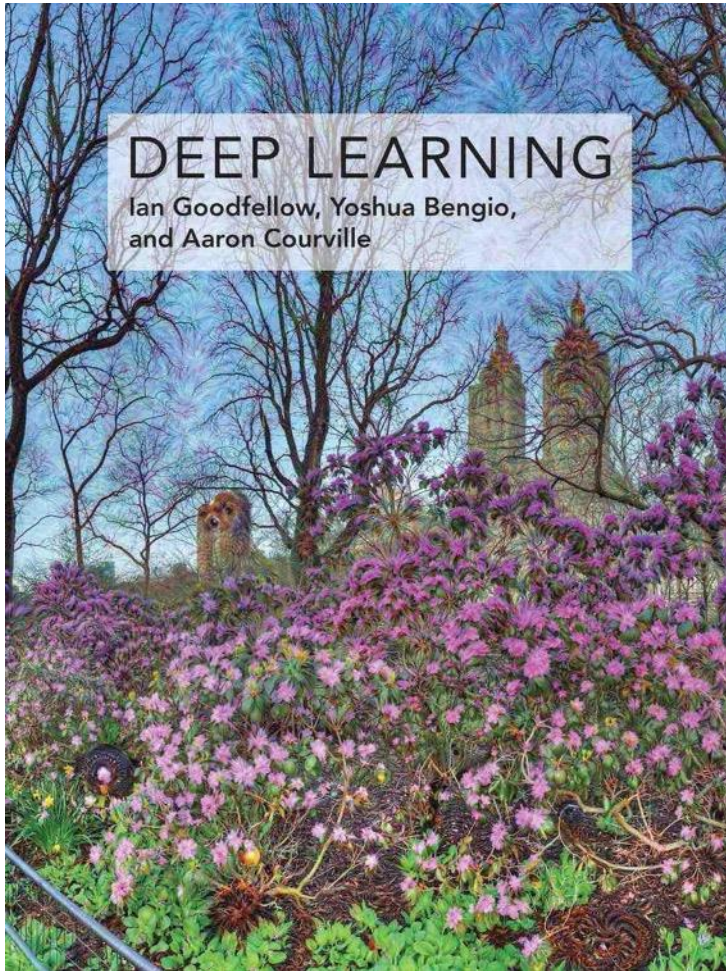
Water-Cooled

9. POWER

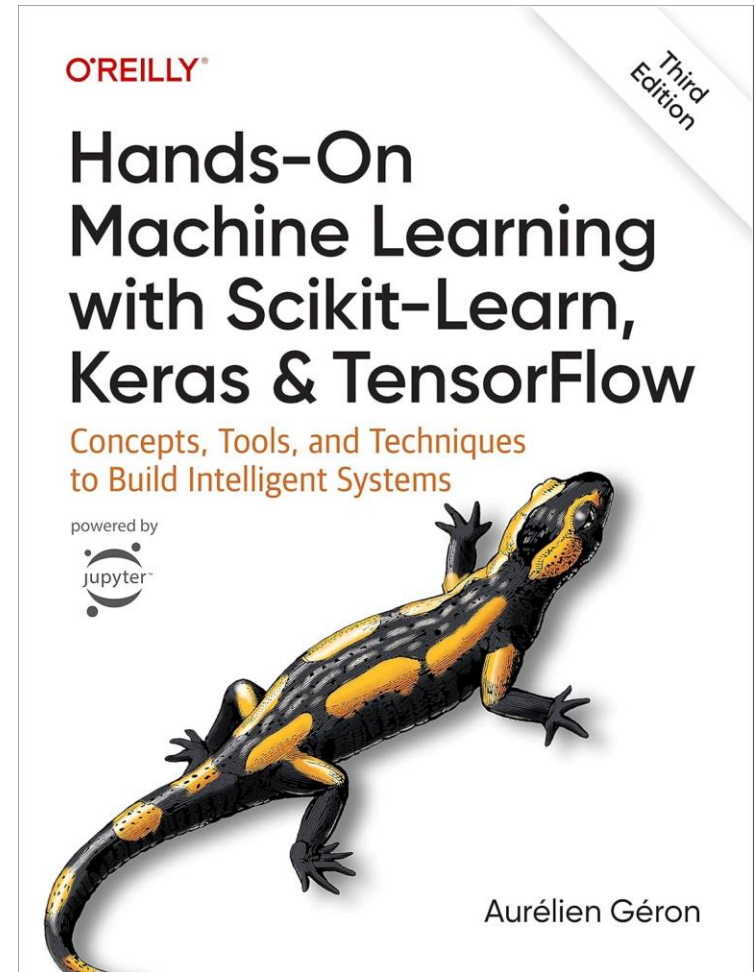
1500 W



Literatura – Duboko učenje



<https://www.deeplearningbook.org>

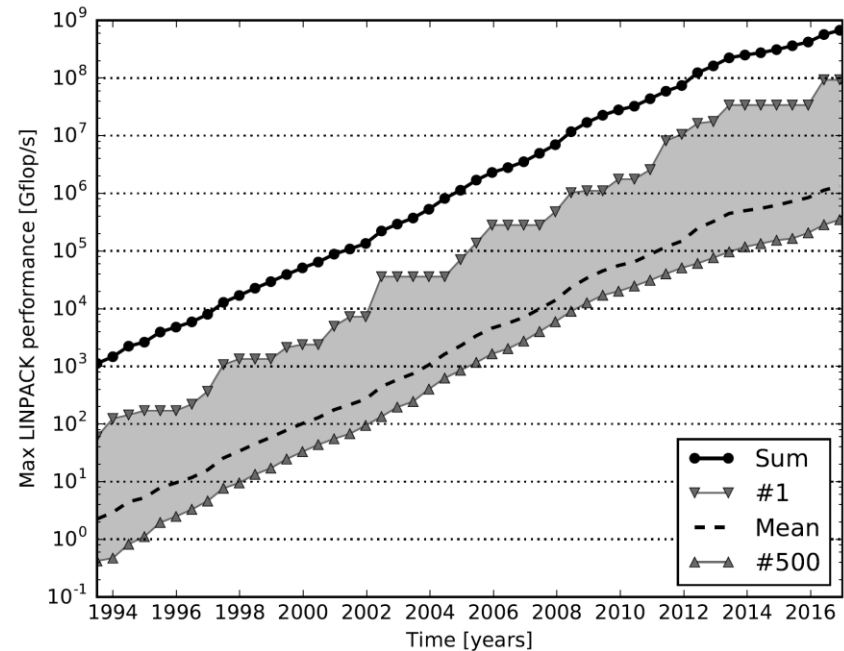


<https://www.oreilly.com/library/view/hands-on-machine-learning/9781098125967/>

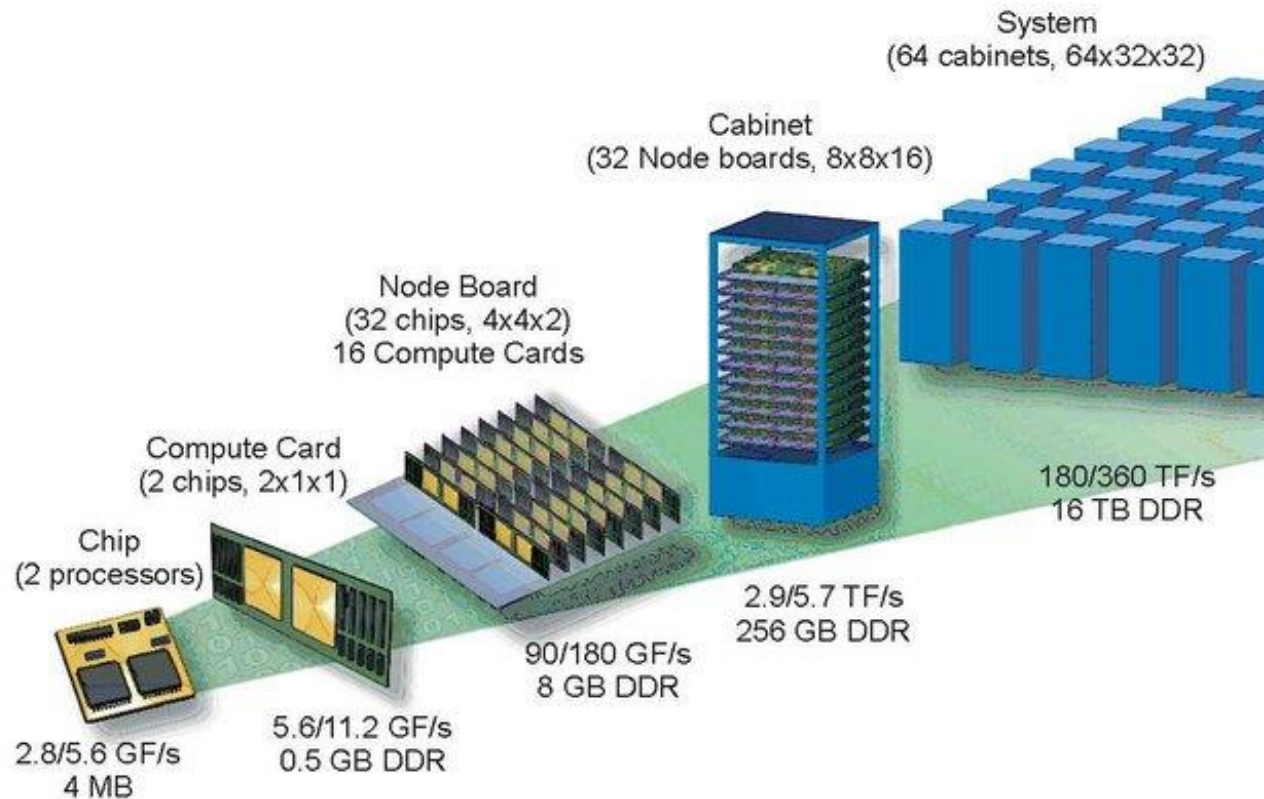
Super-računari i računarstvo visokih performansi

Super-računari – Top500

- **Top500** – super-računari rangirani prema maksimalnim postignutim LINPACK performansama (www.top500.org)
 - Mere se performanse prilikom rešavanja “gustog” sistema linearnih jednačina ($A \cdot x = b$) i iskazuju u Flop/s
 - Na vrhu liste u 2025: *El Capitan* sadrži 11 miliona jezgara (CPU – [AMD Epyc](http://www.amd.com/en-us/processors/epyc), GPU – [AMD Instinct](http://www.amd.com/en-us/processors/instinct)) i postiže 1.74 exaflops
- **Green500** (<http://www.top500.org/green500>): Flop/s-po-vatu postignutih LINPACK performansi



Super-računari – IBM Blue Gene/L



IBM Blue Gene/L super-računar, 65536 čvorova u 64 x 32 x 32 3D torusu

Izvor: <https://homes.cs.washington.edu/~kstrauss/publications/euopar2003.pdf>

Super-računari – IBM Blue Gene/P



IBM Blue Gene/P super-računar Intrepid, Argonne National Lab,
164k CPU jezgara, 40 rekova/kabineta, 3D torus sprežna mreža

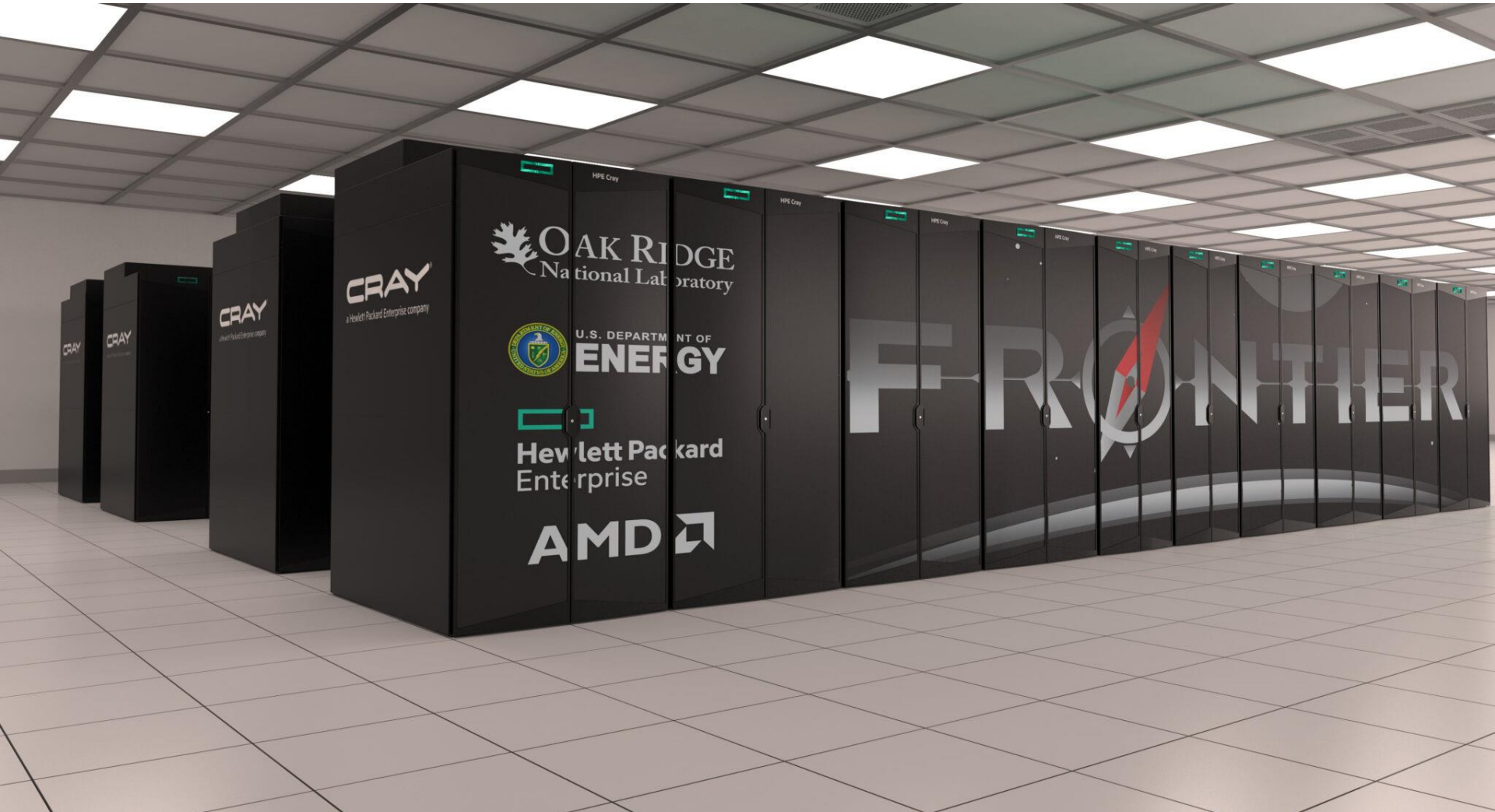
Izvor: [https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

Super-računari – summit



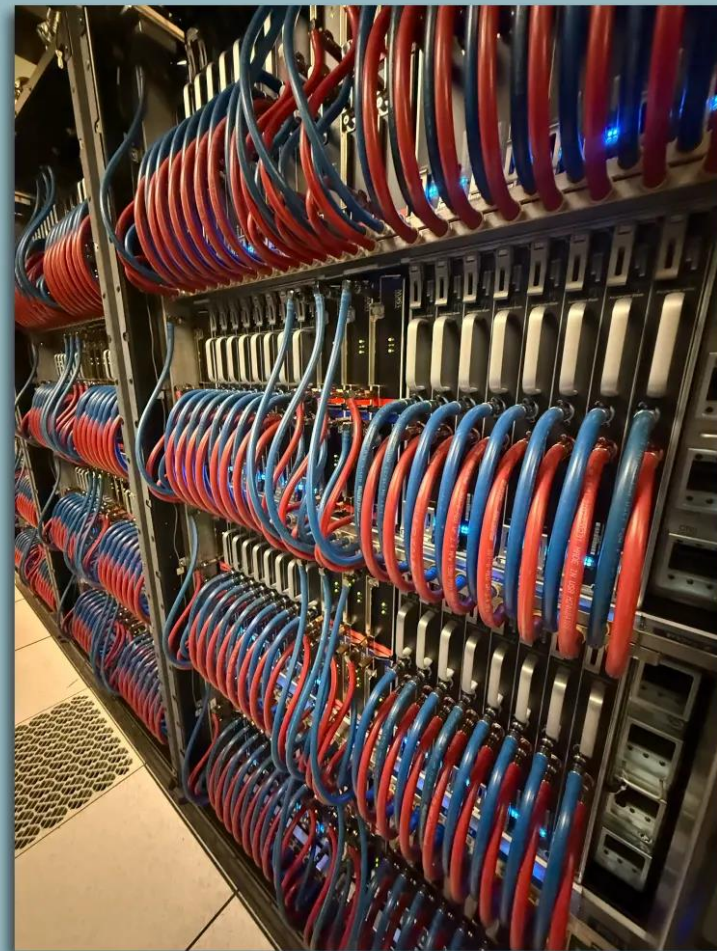
Izvor: [https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

Super-računari – Frontier HPE Cray EX235a



Izvor: <https://en.wikipedia.org/wiki/TOP500>

Super-računari – El Capitan HPE Cray EX255a



Izvor: <https://en.wikipedia.org/wiki/TOP500>



Super-računari – Top500

Rank (previous)	Rmax Rpeak (PetaFLOPS)	Name	Model	CPU cores	Accelerator (e.g. GPU) cores	Total cores (CPUs + accelerators)	Interconnect	Manufacturer	Site country	Year	Operating system
1	1,742.00 2,746.38	El Capitan	HPE Cray EX255a	1,051,392 (43,808 × 24-core Optimized 4th Generation EPYC 24C @1.8 GHz)	9,988,224 (43,808 × 228 AMD Instinct MI300A)	11,039,616	Slingshot-11	HPE	Lawrence Livermore National Laboratory United States	2024	Linux (TOSS)
2	1,353.00 2,055.72	Frontier	HPE Cray EX235a	614,656 (9,604 × 64-core Optimized 3rd Generation EPYC 64C @2.0 GHz)	8,451,520 (38,416 × 220 AMD Instinct MI250X)	9,066,176	Slingshot-11	HPE	Oak Ridge National Laboratory United States	2022	Linux (HPE Cray OS)
3	1,012.00 1,980.01	Aurora	HPE Cray EX	1,104,896 (21,248 × 52-core Intel Xeon Max 9470 @2.4 GHz)	8,159,232 (63,744 × 128 Intel Max 1550)	9,264,128	Slingshot-11	HPE	Argonne National Laboratory United States	2023	Linux (SUSE Linux Enterprise Server 15 SP4)
4 NEW	793.40 930.00	JUPITER	BullSequana XH3000	1,694,592 (23,536 × 72-Arm Neoverse V2 cores Nvidia Grace @3 GHz)	3,106,752 (23,536 × 132 Nvidia Hopper H100)	4,801,344	Quad-rail NVIDIA NDR200 Infiniband	Atos	EuroHPC JU European Union, Jülich, Germany	2025	Linux (RHEL)
5 ▼	561.20 846.84	Eagle	Microsoft NDv5	172,800 (3,600 × 48-core Intel Xeon Platinum 8480C @2.0 GHz)	1,900,800 (14,400 × 132 Nvidia Hopper H100)	2,073,600	NVIDIA Infiniband NDR	Microsoft	Microsoft United States	2023	Linux (Ubuntu 22.04 LTS)

Izvor: <https://en.wikipedia.org/wiki/TOP500>



Super-računari – Top500

Distribution of supercomputers in the TOP500 list by country (as of November 2023)^[30]

Country or Territory	Systems
United States	161
China	104
Germany	36
Japan	32
France	23
United Kingdom	15
Italy	12
South Korea	12
Canada	10
Netherlands	10
Brazil	9
Russia	7
Saudi Arabia	7
Sweden	6
Australia	6
Taiwan	5
Norway	5
India	4
Ireland	4
Poland	4
Finland	3
Spain	3
Singapore	3
Switzerland	3
Czechia	2
Slovenia	2
Austria	2
Bulgaria	2
Luxembourg	2
Thailand	1

novembar 2023

Distribution of supercomputers in the TOP500 list by country (as of June 2025)^[3]

Country or territory	Number of systems
United States	175
European Union	137
China	47
Germany	41
Japan	39
France	25
Italy	17
South Korea	15
Canada	13
United Kingdom	13
Brazil	9
Norway	9
Sweden	9
Taiwan	8
Poland	7
Netherlands	7
Russia	6
Saudi Arabia	6
India	6
Singapore	5
Australia	4
United Arab Emirates	4
Switzerland	4
Czechia	3
Spain	3
Finland	3
Thailand	2
Ireland	2
Slovenia	2

jun 2025

Izvor: <https://en.wikipedia.org/wiki/TOP500>

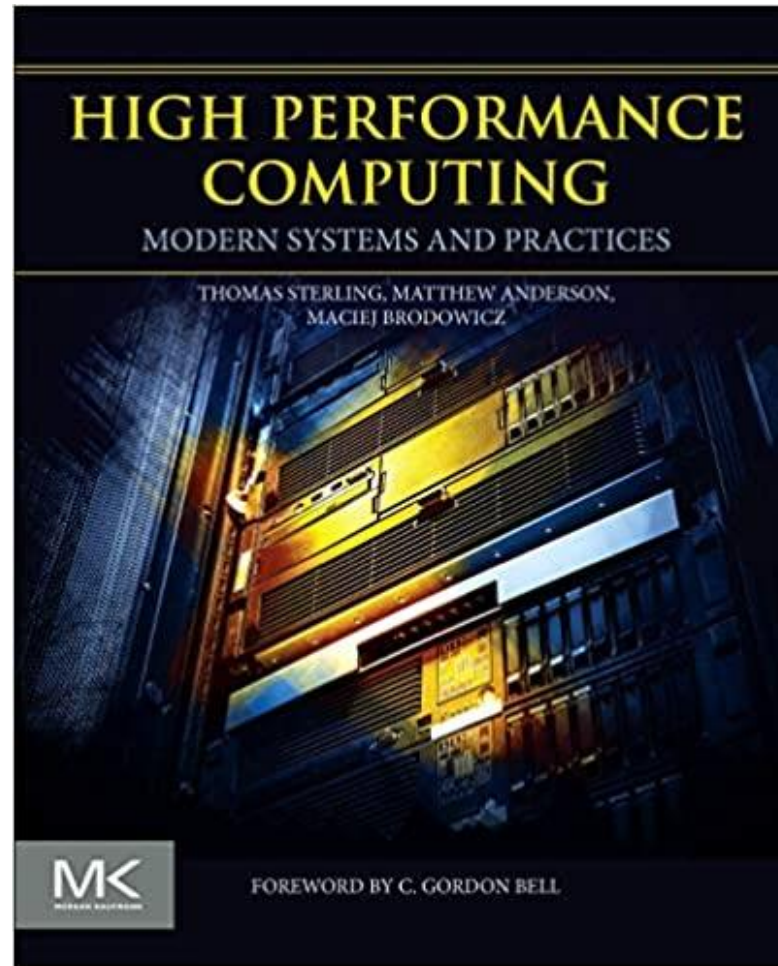


Super-računari – Green500

Rank ↕	Performance per watt ↕ (GFLOPS/watt)	Name	Model Processors, GPU, Interconnect ↕	Vendor ↕	Site Country, year	Rmax (PFLOPS) ↕
1	72.733	JEDI	BullSequana XH3000 Grace Hopper Superchip 72C 3GHz, Nvidia GH200 Superchip, Quad-Rail NVIDIA, InfiniBand NDR200	ParTec ^[14] /EVIDEN (ex-Atos)	EuroHPC/FZJ, Germany, 2024	4.50
2	70.912	ROMEO- 2025	BullSequana XH3000 Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA, InfiniBand NDR200	EVIDEN (ex-Atos)	ROMEO HPC Center ↗, University of Reims Champagne-Ardenne France, 2024	9.86
3	69.098	Adastra 2	HPE Cray EX255a AMD Optimized 4th Generation EPYC 28C 1.8 GHz, AMD Instinct MI300A, Slingshot-11	Hewlett Packard Enterprise	Grand Equipement National de Calcul Intensif - Centre Informatique National de l'Enseignement Supérieur (GENCI-CINES), France, 2024	2.53
4	68.835	Isambard- AI phase 1	HPE Cray EX254n NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11	Hewlett Packard Enterprise	University of Bristol, United Kingdom, 2024	7.42
5	68.177	Otus (GPU only)	ThinkSystem SD665-N V3 AMD EPYC 9655 96C 2.6GHz, NVIDIA H100 SXM5 80GB, Infiniband NDR	Lenovo	Paderborn University PC2, Germany, 2025	4.66

Izvor: www.top500.org/green500

Literatura – HPC



<https://www.elsevier.com/books/high-performance-computing/sterling/978-0-12-420158-3>